

Operations

Week 2 – Live Class

Gerardo Berbeglia
Associate Professor
Melbourne Business School



PROCESS FLOW DIAGRAM:

Components



Storage areas or queues



Tasks or Operations



Flow of material or customers

THREE MEASURES OF PROCESS PERFORMANCE

Cycle Time (time/unit)

- Average time between the output of successive units (e.g. 5 min/unit)
- **Flow rate or throughput rate** is the inverse of the cycle time (e.g. $1/5$ units/min = 0.2 units/min)

Flow time or manufacturing lead time or service lead time

- The average amount of time it takes a unit to get through the process (includes waiting time).

Inventory or Work in process (WIP)

- Average number of units in the process at any point in time (includes the units waiting to be processed).

KEY STEPS IN PROCESS ANALYSIS

1. Choose the process boundaries and the flow unit.
2. Understand how the physical process works and draw a process flow diagram. Show process steps, inventory holding points, and arrows depicting the flow of the product.
3. Determine the capacity of each step in the process expressed as the number of flow units of product that can be processed per unit time.
4. Identify the bottleneck. This is the step with least capacity.
5. Once the bottleneck is identified, think about how the bottleneck influences other process steps. Calculate different performance measures such as the process capacity, flow time, work in process inventory, and labor utilization.
6. Consider changes to improve system performance.

Kristen Cookies

Market context:

- Customised cookies
- Completely fresh
- This requires a make-to-order system

Kristen Cookies

Process flow diagram: One dozen orders

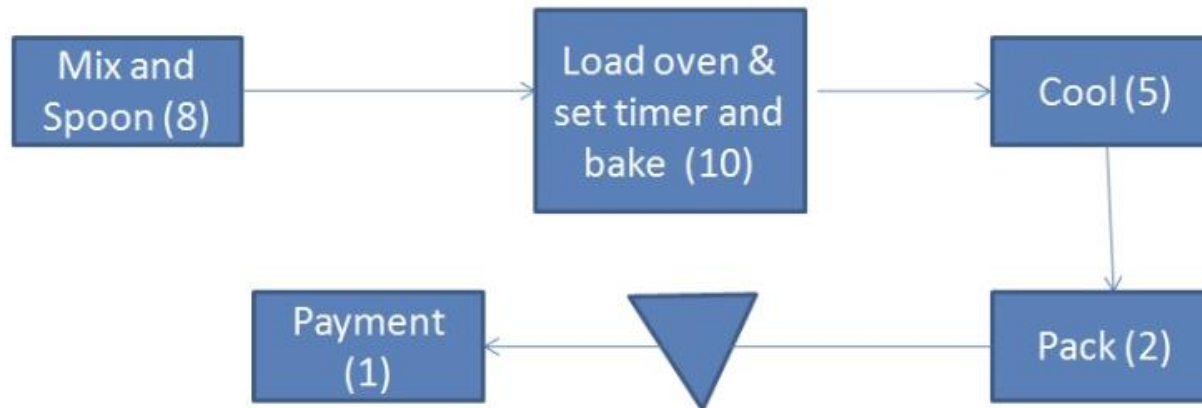


Figure: Kristen's cookies flow Diagram (duration is in minutes)

Bottleneck: Loading oven, set timer and baking (10 minutes)

Kristen Cookies

Discussion:

Should you give any discounts for people who order two dozen cookies, three dozen cookies, or more? If so, how much?

Kristen Cookies

Discussion:

Should you give any discounts for people who order two dozen cookies, three dozen cookies, or more? If so, how much?

Yes



No



Depends on demand



Kristen Cookies

What is the effect of adding another oven?

How much would be the increase in capacity? (imagine orders are for 1 dozen cookies)

How can you look quickly at your order board and tell a new customer when their order will be ready?

LITTLE'S LAW

Applies to queuing systems: arrivals -> items in queue
and items in service -> departures

It connects the three performance measures.

$$\text{Average flow rate (units/time)} = \frac{\text{Average inventory (units)}}{\text{average flow time (time)}}$$

Or in other words, $I = R \times T$

$$\text{Average inventory} = \text{Average flow rate} \times \text{average flow time}$$

Little's Law



1961 result discovered by Prof. John D. C. Little.

- (It is a theorem rather than a law!)

Applies to queuing systems: arrivals -> items in queue and items in service -> departures

Only condition required: the system must be stationary (or in steady state)

No conditions on

- Number of servers cases
- Service times durations (can be stochastic)
- Arrival rates (can be stochastic)
- Number of queues
- Queuing policy (FIFO, LIFO, random)

Examples of flow time, flow rate and inventory

	U.S. immigration department	MBA program	PC manufact- urer	Champagne industry
Flow unit	Application for immigration benefit	MBA student	Computers	Bottles of champagne
Flow rate (R)	Approved or rejected cases: 6.3 million per year	600 students per year	5,000 units/day	260 million bottles per year
Flow time (T)	Average processing time: 7.6 months	2 years	10 days	Average time in cellar: 3.46 years
Inventory (I)	Pending cases: 4 million cases	1,200 students	50,000 computers	900 million bottles

Little's law in the Housing Market



A real estate agent estimates that it takes on average 120 days to sell a house.

You observe that the number of classified ads has ranged between 20 and 30 with an average of 25.

How can you determine the average number of houses sold per year?



Little's law in the Housing Market

Solution:

A home enters the system (or the “process”) when it is listed for sale.

While a home is on the market, it is part of the active system.

A home leaves the system when it is sold.

The time on the market is equivalent to the flow time.

Little's law in the Housing Market



Solution:

Average Flow time = 120 days

Average Inventory = 25 houses

Average flow rate = Average Inventory / Average Flow time
= $25/120 \approx 0.20833$ houses per day = **76 houses per year.**

Little's law in the Housing Market

Activity:

Browse to realestate.com.au and use Little's law to estimate the time it takes to sell an apartment of 1 bedroom in Carlton, Victoria.

Hint: Think how to use the search filters and the properties previously sold.





Little's law in the Housing Market

Activity:

On average, the time taken (in weeks) to sell a 1-bedroom apartment in Carlton, Victoria is approximately

Less than 3	4 to 8	7 to 16	16 to 24	24 to 32	32 or more
					

APPLICATION OF LITTLE'S LAW IN A HOSPITAL: compute patient's flow time.

Estimate the work-in-progress in the system

- Observe the inventory of patients at a couple of random times during the day, giving you an average inventory.
- Let's say this number is 7 patients: four in the waiting room, two already changed and waiting in front of the procedure room, and one in the procedure room.

Determine the throughput per day

- Count the procedure slips or any other records showing how many patients were treated that day. This is the day's output.
- Let's say there were 60 patients over a period of 8 hours. We could say that we have a Throughput rate of $60/8 = 7.5$ patients/hour.

Compute manufacturing lead time or service lead time by applying Little's Law

- Use Little's Law to compute service lead time = work in process
 $\text{Inventory}/\text{Throughput rate} = 7/7.5 = 0.933 \text{ hour} = 56 \text{ minutes}.$
- This tells us that, on average, it takes 56 minutes from the time a patient enters the radiology unit to the time his or her chest X-ray is completed. This gives you an idea about the type of service you are providing and the responsiveness it delivers.

Little's Law and Inventory turns

$$\begin{aligned}\text{Average flow time} &= \text{Inventory} / \text{Flow rate} \\ &= \text{Inventory} / \text{COGS per year}\end{aligned}$$

Average flow time: number of days (or years, or months) it takes to translate a dollar from investment (inventory) to a dollar in revenue.

The inverse is called Inventory turns.

- $\text{Inventory turns} = 1 / \text{flow time}$

Higher inventory turns are desired. Why?

per-unit Inventory cost

Inventory cost calculation

The percentage of the cost associated to inventory (per unit) can be calculated as

$$\frac{\text{Annual inventory costs (expressed as \%)}}{\text{Inventory turns}}$$

Challenges

How do we calculate the inventory costs?

Example:

- Annual inventory costs = 30%
- Inventory turns = 6

$$\text{Per unit Inventory costs (expressed as a \%)} = \frac{30\% \text{ per year}}{6 \text{ turns per year}} = 5\%$$

Mt Kinley Consulting

Mt. Kinley Consulting) Mt. Kinley is a consulting firm that divides its consultants into three classes: associates, managers, and partners. The firm has been stable in size for the last 20 years. Specifically, there have been—and are expected to be—200 associates, 60 managers, and 20 partners.

After four years of working as an associate, a consultant goes "either up or out"; that is, becomes a manager or is dismissed from the company. Similarly, after six years, a manager either becomes a partner or is dismissed. The company recruits MBAs as associate consultants; no hires are made at the manager or partner level. A partner stays with the company for another 10 years (a total of 20 years with the company).

- a. How many new MBA graduates does Mt. Kinley have to hire every year?
- b. What are the odds that a new hire at Mt. Kinley will become partner (as opposed to being dismissed after 4 years or 10 years)

Key Takeaways

We applied process analysis to solve Kristen cookies case

Remarkable connection between three performance measures of any process:
Average inventory, average flow rate and average flow time. [Little's Law]

Application of Little's law to measure inventory turns and average flow time of inventory.

We learnt how to calculate the impact of inventory turns in the per unit inventory cost of a firm.

Connect with us



@MelbBSchool



/MBScarleton



/MelbourneBusinessSchool



@MelbourneBSchool



Melbourne-business-school